Goddard Procedural Requirements (GPR)

DIRECTIVE NO.	GPR 7120.7B	**APPROVED BY Signature:**	*Original Signed By*
EFFECTIVE DATE:	September 17, 2018	**NAME:**	David F. Mitchell
EXPIRATION DATE:	September 17, 2024	**TITLE:**	Director of Flight Projects

COMPLIANCE IS MANDATORY

Responsible Office: Code 400 / Flight Projects Directorate

Title: Funded Schedule Margin and Budget Margin for Flight Projects

PREFACE

P.1 PURPOSE

This GPR documents the funded schedule margin and budget margin requirements for formulating, executing and operating projects at NASA Goddard Space Flight Center (GSFC) that report directly to programs or non-GSFC projects. It describes how those margins are to be tracked and reported during the formulation, implementation and operations phases. The requirements herein represent decades of GSFC's experience in fulfilling NASA Procedural Requirements (NPR) 7120.5, NASA Space Flight Program and Project Management Requirements relevant to budget and schedule, through the management of numerous successful spaceflight projects. It is through the study and analysis of these past missions that these requirements were derived.

P.2 APPLICABILITY

a. This GPR applies to all GSFC-managed space flight projects and enterprise ground system projects that report directly to programs or non-GSFC projects and are developed under NPR 7120.5. Figure 1 illustrates the relationship between programs and projects relative to GPR applicability.

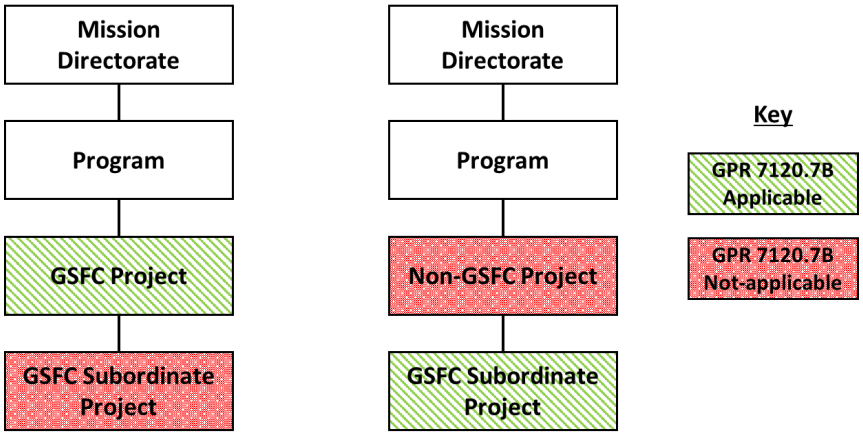

Figure 1. GPR 7120.7B Program and Project Applicability Relationships

b. This GPR does not apply to in-house instrument/payload flight projects that do not report directly to programs and that are subordinate to GSFC flight projects: e.g., TIRS-2, OCI, or L'Ralph. Funded schedule and budget margins for such subordinate projects will be based on the requirements established with their parent mission flight projects.

c. Some projects were begun and implemented under previous versions of this GPR. Appendix D, GPR Applicability Table, shows the applicability by project to the various revisions of the GPR. This GPR is optional for projects developed under NPR 7120.8 and do no harm projects, as defined by GPR 8705.4, Risk Classification and Risk-Based SMA Practices for GSFC Payloads and Systems.

d. In this directive, all document citations are assumed to be the latest version unless otherwise noted.

e. In this directive, all mandatory actions (i.e., requirements) are denoted by statements containing the term "shall." The terms "may" or "can" denote discretionary privilege or permission; "should" denotes a good practice and is recommended but not required; "will" denotes expected outcome; and "are/is" denotes descriptive material.

P.3 AUTHORITY

NPR 7120.5, NASA Space Flight Program and Project Management Requirements

P.4 APPLICABLE DOCUMENTS AND FORMS

NPR 7120.8, NASA Research and Technology Program and Project Management Requirements

GPR 8705.4, Risk Classification and Risk-Based SMA Practices for GSFC Payloads and Systems

P.5 CANCELLATION

GPR 7120.7A, Schedule and Budget Margins for Flight Projects

P.6 SAFETY

None

P.7 TRAINING

None

P.8 RECORDS

Record Title	Record Custodian	Retention
Monthly Status Reviews (MSR) Project tag-up Project budget/schedule records	FPD librarian (FPD Operations hub) FPD librarian (FPD Operations hub) Performing organization	a. Projects of historical significance (see NRRS 8/101 for description): *NRRS 8/103: Temporary. Cut off records at close of program/project or in 5-year blocks. Destroy/delete between 0 and 30 years. b. Programs/projects that do not meet the criteria stated in 8/101: *NRRS 8/107: Temporary. Destroy/delete between 0 and 30 years after program/project termination.

**NRRS 1441.1 – NASA Records Retention Schedules*

P.9 MEASUREMENT/VERIFICATION

Project funded schedule margin and budget margin metrics are reported monthly to the Center Management Council (CMC), which is responsible for ensuring conformance to these requirements.

PROCEDURES

1.0 GENERAL

This document provides guidelines and requirements for the establishment and tracking of funded schedule margin and budget margin, and trends, during the formulation phase of the project, and for the tracking of margin trends during the formulation, implementation and operations phases. In the event of a project re-plan or re-baseline, these requirements will apply.

1.1 Requirements Identification

In this document, requirements are identified by a prefix denoting the directive number followed by a unique number (e.g., **7120.7-R012**). In the interest of readability, the prefix for requirement identifiers have not been included in the body of this document. Use the full identifier when referencing or tracing these requirements in other documents.

Appendix C contains a complete list of the requirements and their locations. In the appendix, requirements are identified as follows:

NEW:	New in this version or document
REVISED:	Changed since the last approved version
CANCELED:	Rescinded
WITHDRAWN:	Absorbed into another requirement or included in the body of the document

1.2 Funded Schedule Margin and Budget Margin Definitions

Funded Schedule Margin: Funded schedule margin is the project-held priced time allowance in working days of duration allocated to project schedules to protect them from uncertainty and risk, while increasing the probability of finishing on time. Funded schedule margin is expressed in both working days and dollars, but does not constitute specific project work scope. With respect to this GPR, funded schedule margin is always considered in relation to the project's critical path. (See Section 2.0 for more on funded schedule margin.)

Budget Margin: Budget margin is the percentage calculated by dividing the available project-held unallocated future expenses (UFE) by the remaining cost-to-go for a designated project phase (minus the UFE as illustrated in Figure 2). Similar to funded schedule margin, UFE is included in project budgets to protect them from uncertainty and risk. (See Section 3.0 for more on budget margin.)

$$\text{Budget* Margin (percentage)} = \frac{\text{The remaining unencumbered, unliened project-held UFE for the relevant phases (Phases A – D or E – F)}}{\text{The estimated remaining cost-to-go for the relevant project phases (Phases A – D or E – F);}} \times 100\%$$

- Includes: encumbrances, liens, and funded schedule margin
- Excludes: project-held UFE (i.e., the numerator of this fraction); education and public outreach and program-managed launch services

*Previously called financial contingency or cost reserve, typically expressed as a percentage

Figure 2. Budget Margin Formula

1.3 Funded Schedule Margin and Budget Margin Requirements by Project Type

Funded schedule margin and budget margin requirements for mission flight projects, standalone instrument/payload flight projects, and enterprise ground system projects are summarized in Tables 1, 2, and 3 respectively, and described in further detail in subsequent sections.

1.4 Relationship to Earned Value Management

Although the application of funded schedule margin and budget margin directly impacts a project's baseline and earned value management (EVM) system (EVMS), it is not the intent of this directive to serve as a guide or instruction on how to incorporate funded schedule margin or budget margins into a project baseline or EVMS. Projects should address the application of funded schedule margin and budget margin in their EVM implementation plans.

1.5 Noncompliance with Funded Schedule Margin and Budget Margin Requirements

The CMC will direct actions in response to noncompliance with the required funded schedule margin and budget margin levels defined herein. At Key Decision Points (KDP), the position of the CMC will be reflected in GSFC's letter to the KDP decision authority.

Table 1. Funded Schedule Margin and Budget Margin Requirements for Mission Flight Projects

Event	GPR Require -ment	Margin Type	Phase A	Phase B	Phase C	Phase D	Phase E	Phase F
At Key Decision Point (KDP) A	R003 R004 R005	Funded Schedule Margin (along the critical path)	None	None	Interval 1: 1.5 months funded schedule margin allocated for each 12 months of planned duration between KDP-C approval and start of Observatory I&T *(see Figure 3)*	Interval 2: 2.0 months funded schedule margin allocated for each 12 months of planned duration between start of Observatory I&T and delivery to launch site (or to planned storage) AND Interval 3: 1 week funded schedule margin allocated for each month of planned duration from observatory delivery to launch site to launch *(see Figure 3)*	None	None
	R009	Budget Margin (through encumbrances & liens)	≥ 25% on cost-to-go through Phase D (required) ≥ 30% on cost-to-go through Phase D (goal) *(see Figure 6)*				15% budget margin for Phases E & F *(see Figure 6)*	
At KDP-B	R003 R004 R005	Funded Schedule Margin (along the critical path)	N/A	None	Interval 1: 1.5 months funded schedule margin allocated for each 12 months of planned duration between KDP-C approval and start of Observatory I&T *(see Figure 3)*	Interval 2: 2.0 months funded schedule margin allocated for each 12 months of planned duration between start of Observatory I&T and delivery to launch site (or to planned storage) AND Interval 3: 1 week funded schedule margin allocated for each month of planned duration from observatory delivery to launch site to launch *(see Figure 3)*	None	None
	R009	Budget Margin (through encumbrances & liens)	N/A	≥ 25% on cost-to-go through Phase D (required) ≥ 30% on cost-to-go through Phase D (goal) *(see Figure 6)*			15% budget margin for Phases E & F *(see Figure 6)*	
At KDP-C	R003 R004 R005	Funded Schedule Margin (along the critical path)	N/A	N/A	Interval 1: 1.5 months funded schedule margin allocated for each 12 months of planned duration between KDP-C approval and start of Observatory I&T *(see Figure 3)*	Interval 2: 2.0 months funded schedule margin allocated for each 12 months of planned duration between start of Observatory I&T and delivery to launch site (or to planned storage) AND Interval 3: 1 week funded schedule margin allocated for each month of planned duration from observatory delivery to launch site to launch *(see Figure 3)*	None	None
	R010	Budget Margin (through encumbrances & liens)	N/A	N/A	≥ 25% on cost-to-go through Phase D *(see Figure 6)*		15% budget margin for Phases E & F *(see Figure 6)*	
At KDP-D	R004 R005	Funded Schedule Margin (along the critical path)	N/A	N/A	N/A	Interval 2: 2.0 months funded schedule margin allocated for each 12 months of planned duration between start of Observatory I&T and delivery to launch site (or to planned storage) AND Interval 3: 1 week funded schedule margin allocated for each month of planned duration from observatory delivery to launch site to launch *(see Figure 3)*	None	None
	R011	Budget Margin (through encumbrances & liens)	N/A	N/A	N/A	≥ 20% on cost-to-go through Phase D *(see Figure 6)*	10% budget margin for Phases E & F *(see Figure 6)*	
At Delivery to Launch Site and Thereafter	R005	Funded Schedule Margin (along the critical path)	N/A	N/A	N/A	Interval 3: 1 week funded schedule margin allocated for each month of planned duration from observatory delivery to launch site to launch *(see Figure 3)*	None	None
	R012	Budget Margin (through encumbrances & liens)	N/A	N/A	N/A	≥ 10% on cost-to-go through Phase D *(see Figure 6)*	10% budget margin for Phases E & F *(see Figure 6)*	

Table 2. Funded Schedule Margin and Budget Margin Requirements for Standalone Instrument/Payload Flight Projects

Event	GPR Require -ment	Margin Type	Phase A	Phase B	Phase C	Phase D	Phases E & F
At KDP-A	R017 R018	Funded Schedule Margin (along the critical path)	None	None	Interval 1: 1.5 months funded schedule margin allocated for each 12 months of planned duration between KDP-C approval and start of instrument/payload I&T *(see Figure 4)*	Interval 2: 2.0 months funded schedule margin allocated for each 12 months of planned duration between start of instrument/payload I&T and instrument/payload delivery *(see Figure 4)*	None
	R019	Budget Margin (through encumbrances & liens)	≥ 25% on cost-to-go through Phase D (required) ≥ 30% on cost-to-go through Phase D (goal) *(see Figure 7)*				15% budget margin for Phases E & F *(see Figure 7)*
At KDP-B	R017 R018	Funded Schedule Margin (along the critical path)	N/A	None	Interval 1: 1.5 months funded schedule margin allocated for each 12 months of planned duration between KDP-C approval and start of instrument/payload I&T *(see Figure 4)*	Interval 2: 2.0 months funded schedule margin allocated for each 12 months of planned duration between start of instrument/payload I&T and instrument/payload delivery *(see Figure 4)*	None
	R019	Budget Margin (through encumbrances & liens)	N/A	≥ 25% on cost-to-go through Phase D (required) ≥ 30% on cost-to-go through Phase D (goal) *(see Figure 7)*			15% budget margin for Phases E & F *(see Figure 7)*
At KDP-C	R017 R018	Funded Schedule Margin (along the critical path)	N/A	N/A	Interval 1: 1.5 months funded schedule margin allocated for each 12 months of planned duration between KDP-C approval and start of instrument/ payload I&T *(see Figure 4)*	Interval 2: 2.0 months funded schedule margin allocated for each 12 months of planned duration between start of instrument/ payload I&T and instrument/payload delivery *(see Figure 4)*	None
	R020	Budget Margin (through encumbrances & liens)	N/A	N/A	≥ 25% on cost-to-go through Phase D *(see Figure 7)*		15% budget margin for Phases E & F *(see Figure 7)*
At KDP-D	R018	Funded Schedule Margin (along the critical path)	N/A	N/A	N/A	Interval 2: 2.0 months funded schedule margin allocated for each 12 months of planned duration between start of instrument/ payload I&T and instrument/payload delivery *(see Figure 4)*	None
	R021	Budget Margin (through encumbrances & liens)	N/A	N/A	N/A	≥ 20% on cost-to-go through Phase D *(see Figure 7)*	10% budget margin for Phases E & F *(see Figure 7)*
At Instrument/ Payload Delivery and Thereafter	N/A	Funded Schedule Margin (along the critical path)	N/A	N/A	N/A	None	None
	R022	Budget Margin (through encumbrances & liens)	N/A	N/A	N/A	≥ 10% on cost-to-go through Phase D (see Figure 7)	10% budget margin for Phases E & F *(see Figure 7)*

Table 3. Funded Schedule Margin and Budget Margin Requirements for Enterprise Ground System Projects

Event	GPR Require-ment	Margin Type	Phase A	Phase B	Phase C	Phase D	Phase E	Phase F
At KDP-A **OR** Ground System System Requirements Review (SRR) if no KDP-A	R006	Funded Schedule Margin (along the critical path)	None	None	If KDP-A: 2.0 months of funded schedule margin allocated for each 12 months of planned duration between KDP-C approval and mission-level Operational Readiness Review (ORR), ground system freeze, final acceptance, or other appropriate milestone If no KDP-A: 2.0 months of funded schedule allocated for each 12 months of planned duration between ground system PDR and mission-level ORR, ground system freeze, final acceptance, or other appropriate milestone *(see Figure 5)*		None	None
	R023	Budget Margin (through encumbrances & liens)	≥ 25% on cost-to-go through Phase D (or mission-level ORR, ground system freeze, final acceptance, or other appropriate milestone if no Phase D) (required) ≥ 30% on cost-to-go through Phase D (or mission-level ORR, ground system freeze, final acceptance, or other appropriate milestone if no Phase D) (goal) *(see Figure 8)*				15% budget margin for Phases E & F (or equivalent if no E&F) *(see Figure 8)*	
At KDP-B **OR** Ground System System Design Review (SDR) if no KDP-B	R006	Funded Schedule Margin (along the critical path)	N/A	None	If KDP-B: 2.0 months of funded schedule margin allocated for each 12 months of planned duration between KDP-C approval and mission-level ORR, ground system freeze, or other appropriate milestone If no KDP-B: 2.0 months of funded schedule margin allocated for each 12 months of planned duration between ground system PDR and mission-level ORR, ground system freeze, final acceptance, or other appropriate milestone *(see Figure 5)*		None	None
	R023	Budget Margin (through encumbrances & liens)	N/A	≥ 25% on cost-to-go through Phase D (or mission-level ORR, ground system freeze, final acceptance, or other appropriate milestone if no Phase D) (required) ≥ 30% on cost-to-go through Phase D (or mission-level ORR, ground system freeze, final acceptance, or other appropriate milestone if no Phase D) (goal) *(see Figure 8)*				15% budget margin for Phases E & F (or equivalent if no E&F) *(see Figure 8)*	
At KDP-C **OR** Ground System PDR if no KDP-C	R006	Funded Schedule Margin (along the critical path)	N/A	N/A	If KDP-C: 2.0 months of funded schedule margin allocatged for each 12 months of planned duration between KDP-C approval and mission-level ORR, ground system freeze, final acceptance, or other appropriate milestone If no KDP-C: 2.0 months of funded schedule margin allocated for each 12 months of planned duration between ground system PDR and mission-level ORR, ground system freeze, final acceptance, or other appropriate milestone *(see Figure 5)*		None	None
	R024	Budget Margin (through encumbrances & liens)	N/A	N/A	≥ 25% on cost-to-go through Phase D (or mission-level ORR, ground system freeze, final acceptance, or other appropriate milestone if no Phase D) *(see Figure 8)*		15% budget margin for Phases E & F (or equivalent if no E&F)	

2.0 FUNDED SCHEDULE MARGIN

When planning and tracking funded schedule margin for mission flight, subordinate instrument/payload flight, and enterprise ground system projects, the following considerations apply:

a. Funded schedule margin is the priced time allowance necessary to protect the project schedule against uncertainty and risk.

b. Funded schedule margin requirements in this GPR represent minimum levels relative to the project critical path and actual amounts may be higher based on an assessment of uncertainty and risk.

c. Funded schedule margin consists of both working days of duration and the funds associated with that duration.

d. The required amount of funded schedule margin funds will be calculated using a burn rate sufficient to cover schedule delays for the phase of the project (e.g., average burn rate for the critical path(s) over the appropriate period).

e. Funded schedule margin is not included in project-held UFE.

f. Contractor-held funded schedule margin that is not directly controlled by the NASA project manager is excluded from the project-held funded schedule margin planned and tracked for this GPR.

g. The required amount of funded schedule margin duration will be calculated in working days based on a 5-day work week, which equates to approximately 21 working days in a calendar month. (See Appendix A definition.)

Funded schedule margin requirements unique to mission flight projects, standalone instrument/payload flight projects, and enterprise ground system projects are described in the sections that follow.

2.1 Funded Schedule Margin for Mission Flight Projects

In the planning stages of a mission flight project, including re-plans and re-baselines of applicable mission flight projects, the following minimum amounts of funded schedule margin along the critical path shall be baselined (see Figure 3):

a. Interval 1: From KDP-C approval (i.e., confirmation) to the start of observatory I&T (see Appendix A definition for start of observatory I&T), 1.5 months of funded schedule margin allocated for every 12 months of Interval 1 planned duration. **[R003]**

b. Interval 2: From the start of observatory I&T to delivery of the observatory to the launch site (or to planned storage), 2 months of funded schedule margin allocated for every 12 months of Interval 2 planned duration. **[R004]**

c. Interval 3: From observatory delivery to the launch site to launch, 1 week of funded schedule margin allocated for every month of Interval 3 planned duration. **[R005]**

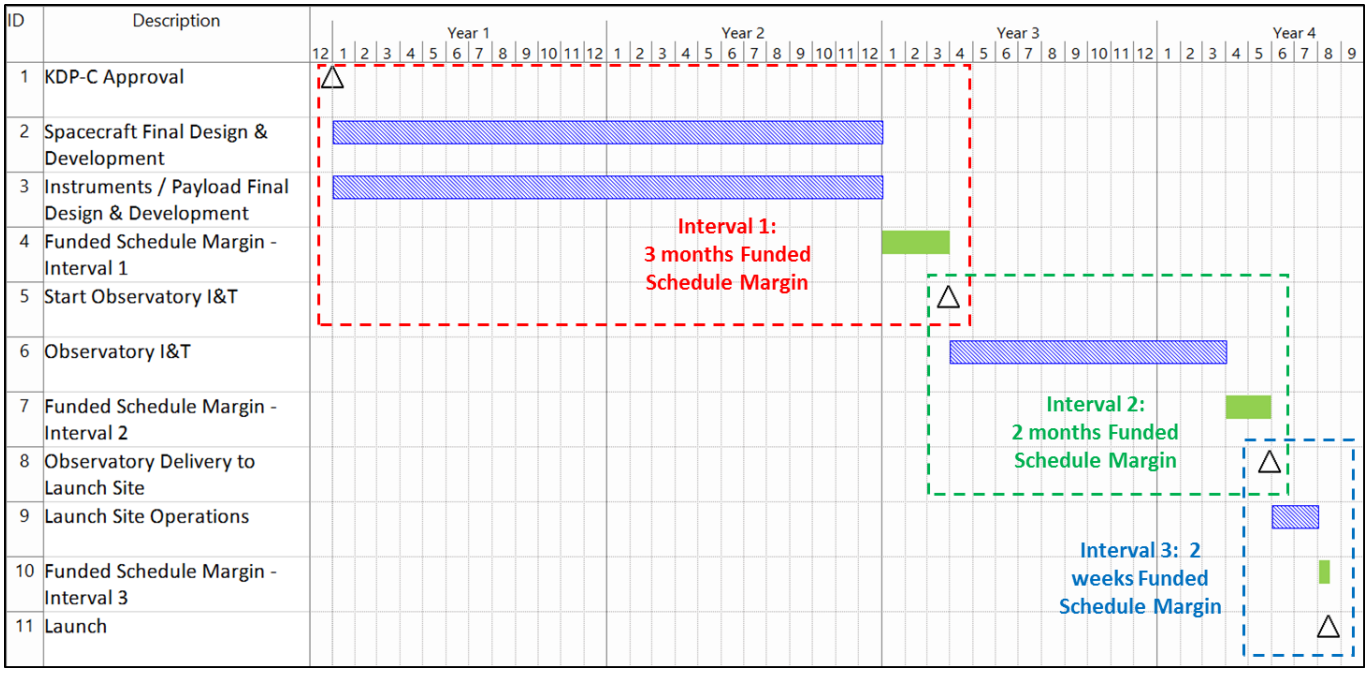

Figure 3. Mission Flight Project Funded Schedule Margin

For subordinate in-house instrument/payload projects delivering to GSFC's mission flight projects, the mission flight project management team will work with the subordinate in-house instrument/payload project management team to determine which organization will be responsible for planning, controlling and reporting subordinate instrument/payload funded schedule margin in accordance with the mission flight project's requirements. This funded schedule margin management approach will be documented in the mission flight project's project plan and the subordinate instrument/payload flight project's instrument project implementation plan. Subordinate in-house instrument/payload project management team(s) should control some portion of funded schedule margin.

2.2 Funded Schedule Margin for Standalone Instrument/Payload Flight Projects

The following minimum funded schedule margin amounts along the critical path shall be baselined in the planning stages of a standalone instrument/payload flight project, including re-plans and re-baselines of applicable standalone instrument/payload flight projects, (see Figure 4):

a. Interval 1: From KDP-C approval to the start of instrument/payload I&T, 1.5 months of funded schedule margin allocated for every 12 months of Interval 1 planned duration. (The start of instrument/payload I&T varies depending on the nature of the project. Therefore, during the planning phase, projects will define an appropriate milestone or event that constitutes the start of instrument/payload level I&T.) **[R017]**

b. Interval 2: From the start of instrument/payload I&T to instrument/payload delivery, 2 months of funded schedule margin allocated for every 12 months of Interval 2 planned duration, where instrument/payload delivery may include physical arrival at the launch site or other facility. **[R018]**

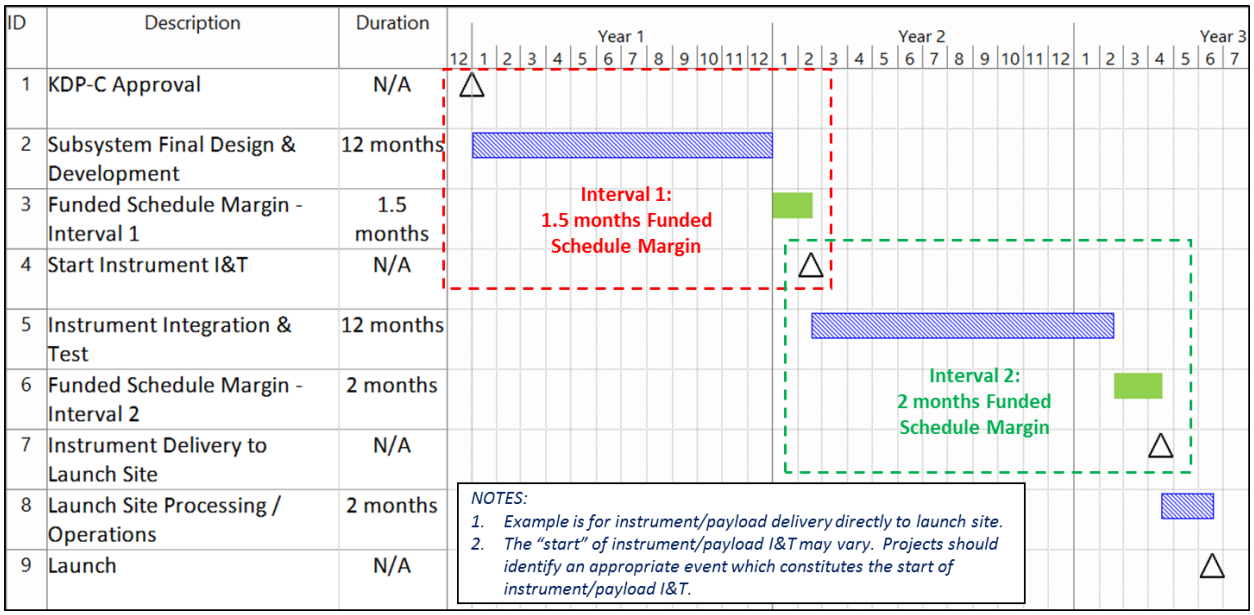

Figure 4. Standalone Instrument/Payload Flight Project Funded Schedule Margin

2.3 Funded Schedule Margin for Enterprise Ground System Projects

An enterprise ground system project is a large-scale ground system developed to support a specific program's requirements. It is independently managed and reports directly to that program. (Subordinate ground systems embedded within a mission are not considered enterprise ground systems.) Since enterprise ground systems' requirements may vary, their project leadership teams will consult with FPD management on specific funded schedule margin allocations. In general, during the planning stages of an enterprise ground system project, including re-plans and re-baselines of applicable enterprise ground system projects, the following minimum amounts of funded schedule margin along the critical path shall be baselined (see Figure 5):

a. From KDP-C approval (or ground system PDR if there is no enterprise ground system project KDP-C) to mission level ORR, ground system freeze, final acceptance, or other appropriate milestone, 2.0 months of funded schedule margin allocated for every 12 months of planned duration. **[R006]**

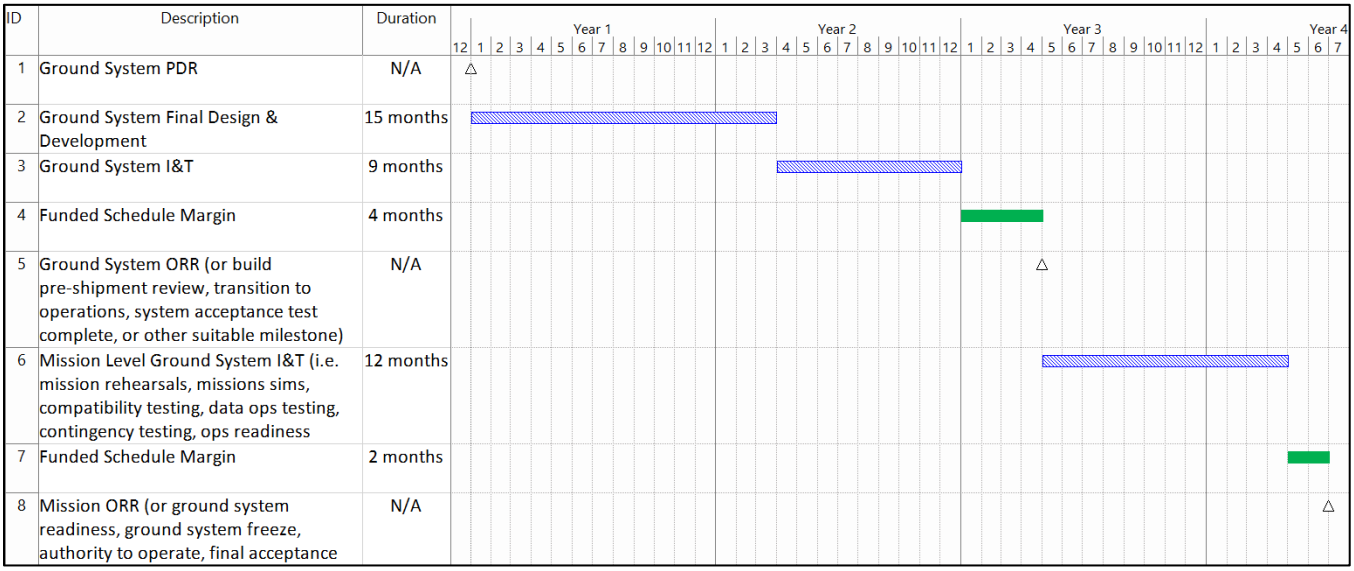

Figure 5. Enterprise Ground System Project Funded Schedule Margin

2.4 Funded Schedule Margin for Other Project Types

This GPR will not apply to some projects and activities (see section P.2), yet these projects should develop an approach to handling the planning and tracking of funded schedule margin. Generally, project managers should derive the required amount of funded schedule margin based on an assessment of the development risk and through consultation with appropriate directorate and center management.

3.0 BUDGET MARGIN

When planning and tracking budget margin for mission flight projects, standalone instrument/payload projects, and enterprise ground system projects, the following considerations apply:

a. Budget margin is the percentage derived from dividing the available project-held UFE by the remaining cost-to-go. (Refer to Figure 2 for the specific elements of this ratio.)

b. The UFE associated with budget margin is the funding allowance necessary to protect the project budget against uncertainty and risk.

c. The budget margin requirements in this GPR represent minimum percentage levels and actual amounts may be higher based on an assessment of uncertainty and risk.

d. Budget margin will not be required for education and public outreach or program-managed launch services.

e. Project-held UFE does not include program-held or NASA Headquarters-held UFE.

f. Projects in Phases E and F do not require budget margin for activities such as grants, guest or guaranteed observer programs, or administrative support.

g. The budget margin percentage level will be maintained from a KDP approval until it "steps down" at the next KDP approval, as illustrated in Figures 6, 7, and 8.

h. Funds for funded schedule margin are not included as part of the project-held UFE.

i. Funds for funded schedule margin will be traceable to funded schedule margin days and identified as a separate budget item within the cost-to-go.

j. Project-held UFE does not include contractor-held budget margin no matter how that margin is defined (e.g., cost reserve, management reserve, financial contingency, etc.).

Budget margin requirements unique to mission flight projects, standalone instrument/payload flight projects, and enterprise ground system projects are described in the sections that follow.

3.1 Budget Margin for Mission Flight Projects

During the life cycle of a mission flight project, including re-plans and rebaselines, the following minimum budget margins, through encumbrances and liens, shall be baselined (see Figure 6):

a. At KDP-A and KDP-B, mission flight projects shall have a minimum budget margin level of 25% with a goal of 30% on remaining cost-to-go through Phase D, and 15% for Phases E and F. **[R009]**

b. At KDP-C, mission flight projects shall have a budget margin level of 25% or higher on remaining cost-to-go through Phase D, and 15% for Phases E and F. **[R010]**

c. At KDP-D, mission flight projects shall have a budget margin level of 20% or higher on remaining cost-to-go through Phase D, and 10% for Phases E and F. **[R011]**

d. At the time of delivery to the launch site and thereafter, mission flight projects shall have a budget margin level of 10% or higher on remaining cost-to-go through Phase D, and 10% for Phases E and F. **[R012]**

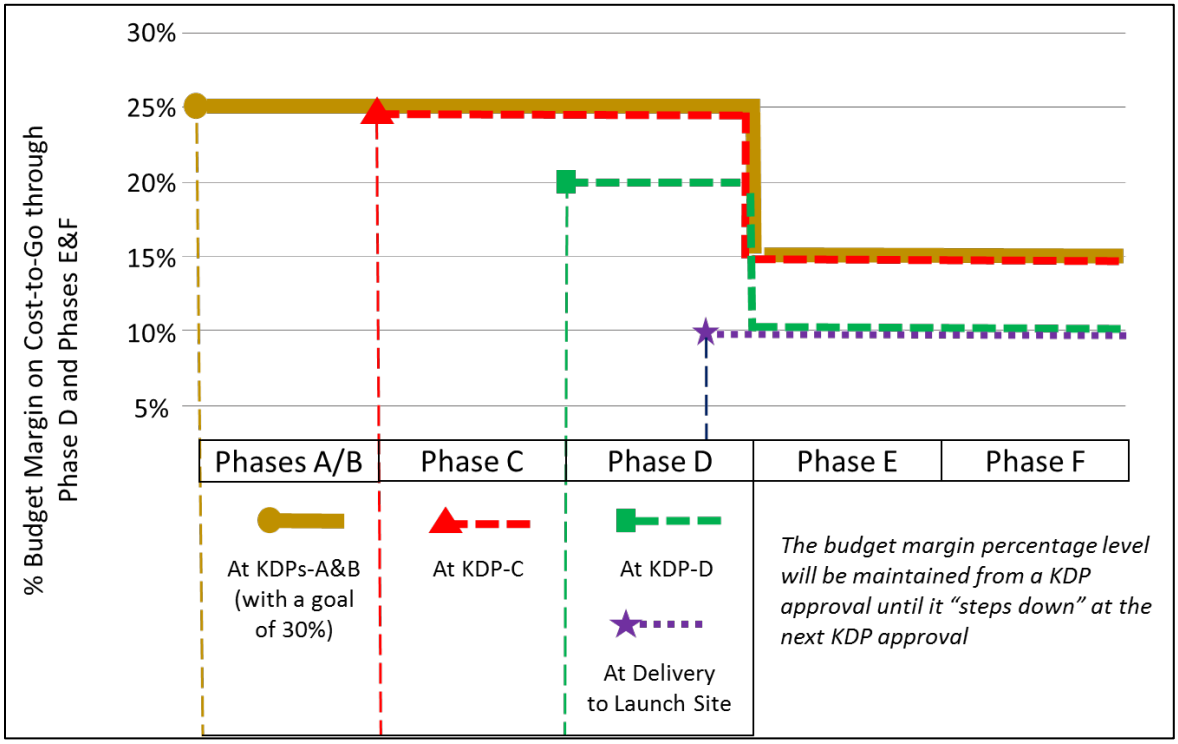

Figure 6. Mission Flight Project Budget Margin

For subordinate in-house instrument/payload projects delivering to GSFC mission flight projects, the mission project management team will work with the subordinate in-house instrument/payload project management team to determine which organization will be responsible for planning, controlling, and reporting the subordinate instrument/payload project's budget margin. A UFE management approach will be documented in the mission flight project's project plan and the subordinate in-house instrument/payload project's instrument project implementation plan. Subordinate in-house instrument/payload project management team(s) should control some portion of project-held UFE.

3.2 Budget Margin for Standalone Instrument/Payload Flight Projects

During the life cycle of a standalone instrument/payload flight project, including re-plans and re-baselines, the following minimum budget margins shall be baselined (see Figure 7):

a. At KDP-A and KDP-B, standalone instrument/payload flight projects shall have a minimum budget margin level of 25% with a goal of 30% on remaining cost-to-go through Phase D, and 15% for Phases E and F. **[R019]**

b. At KDP-C, standalone instrument/payload flight projects shall have a budget margin level of 25% or higher on remaining cost-to-go through Phase D, and 15% for Phases E and F. **[R020]**

c. At KDP-D, standalone instrument/payload flight projects shall have a budget margin level of 20% or higher on remaining cost-to-go through Phase D, and 10% for Phases E and F. **[R021]**

d. At the time of instrument/payload delivery and thereafter, standalone instrument/payload flight projects shall have a budget margin level of 10% or higher on remaining cost-to-go through Phase D, and 10% for Phases E and F. **[R022]**

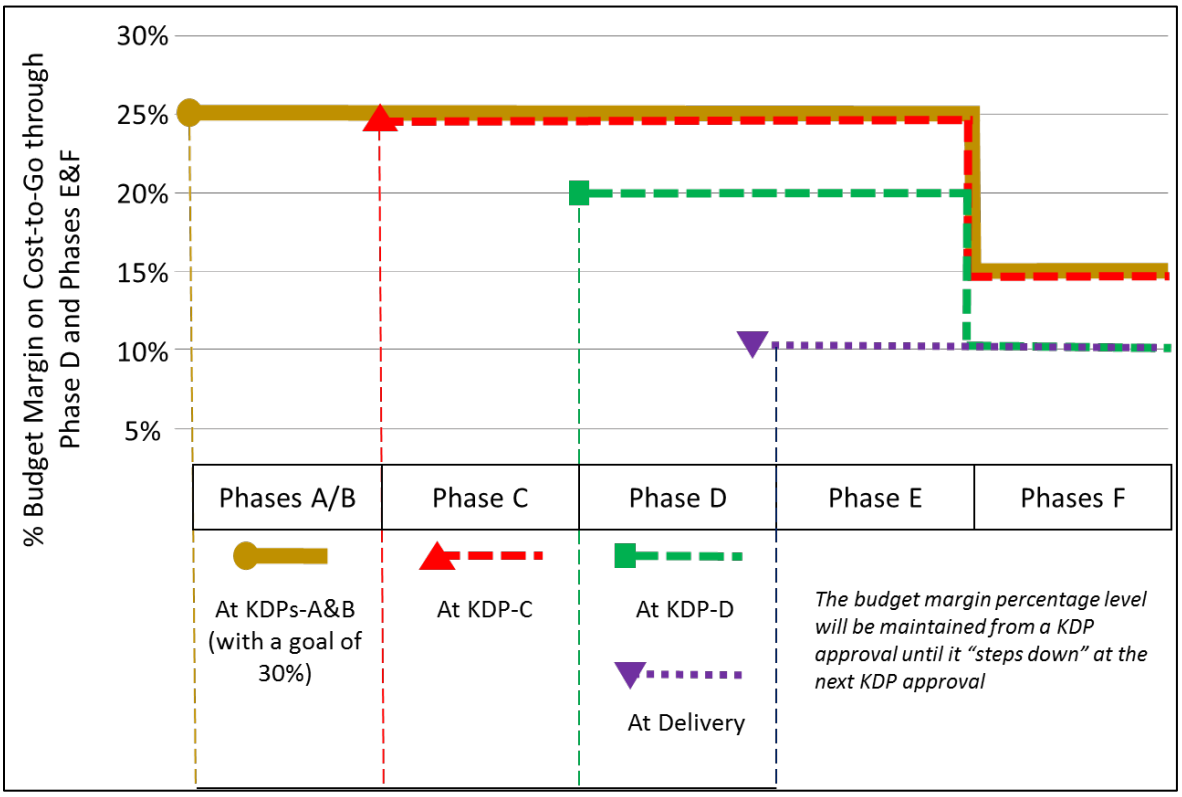

Figure 7. Standalone Instrument/Payload Flight Project Budget Margin

3.3 Budget Margin for Enterprise Ground System Projects

During the life cycle of an enterprise ground system project, including re-plans and re-baselines, the following minimum budget margin shall be baselined (see Figure 8):

a. At KDP-A and KDP-B (or ground system SRR/SDR if there is no enterprise ground system project KDP-A/KDP-B), enterprise ground system projects shall have a minimum budget margin level of 25% with a goal of 30% on remaining cost-to-go through Phase D (or ground system ORR, ground system freeze, or other appropriate milestone if no Phase D), and 15% for Phases E and F (or equivalent if no E&F). **[R023]**

b. At KDP-C (or enterprise ground system PDR if there is no enterprise ground system project KDP-C), enterprise ground system projects shall have a budget margin level of 25% or higher on remaining cost-to-go through Phase D (or ground system ORR, ground system freeze, or other appropriate milestone if no Phase D) and 15% for Phases E and F (or equivalent if no E&F). **[R024]**

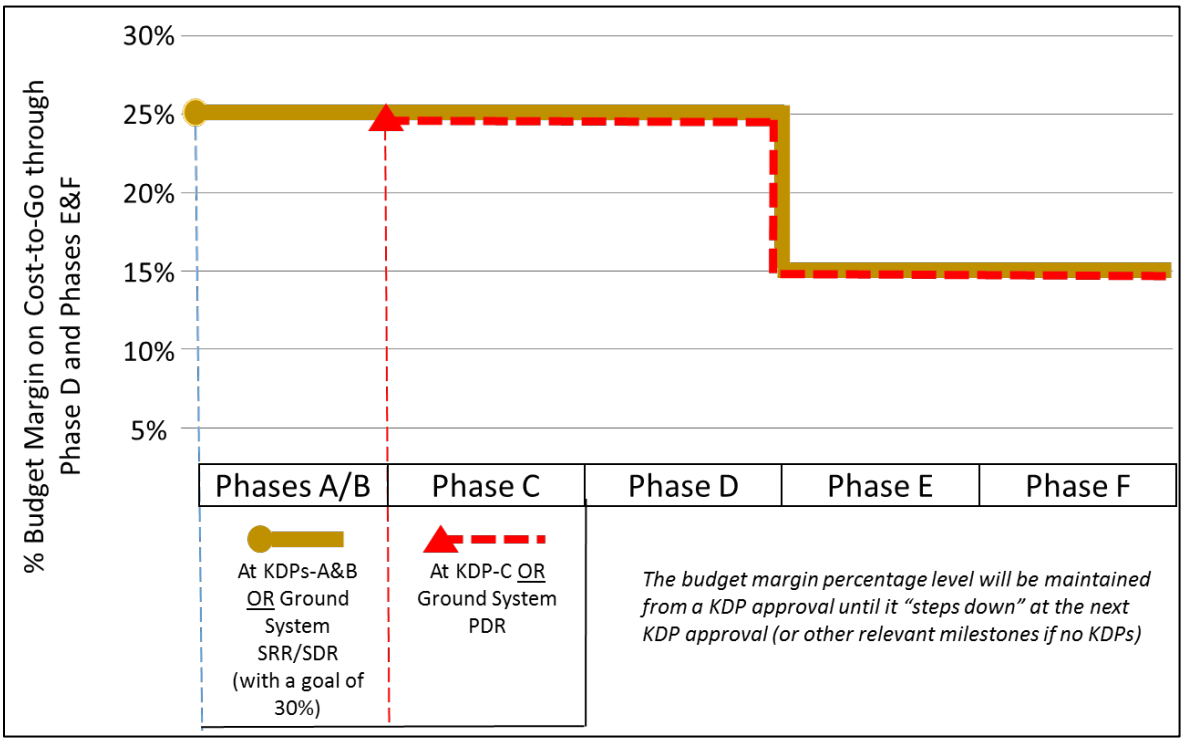

Figure 8. Enterprise Ground System Budget Margin

3.4 Budget Margin for Other Project Types

This GPR will not apply to some projects and activities (see section P.2) yet these projects should develop an approach to handling the planning and tracking of budget margin. Generally, project managers should derive the required amount of budget margin based on an assessment of the development risk and through consultation with appropriate directorate and center management.

4.0 MONTHLY REPORTING

a. At the FPD's monthly project tag-ups and the Center's MSRs, projects will present their funded schedule margin and budget margin status in accordance with the current MSR reporting guidance.

b. The CMC will provide direction to projects that deviate from these requirements.

Appendix A – Definitions

This appendix provides definitions for the terms used in this GPR.

A.1 Budget Margin: The percentage calculated by dividing the remaining unencumbered, unliened project-held UFE by the remaining cost-to-go for a designated project phase minus the UFE. Similar to funded schedule margin, UFE is included in project budgets to protect them from uncertainty and risk.

A.2 Remaining Cost-to-Go: Total estimated cost of all work remaining (to-go) for a project to complete the relevant phases (A-D or E-F), not including program-managed launch services or education and public outreach. Cost-to-go includes all encumbrances, liens, and funded schedule margin.

A.3 Funded Schedule Margin: The project-held priced time allowance in working days of duration allocated to project schedules to protect them from uncertainty and risk, while increasing the probability of finishing on time. Funded schedule margin is expressed in both working days and dollars, but does not constitute specific project work scope. (NOTE: The figure of approximately 21 working days per month is calculated from the total days in one calendar year, minus weekend days, minus the 10 government holidays and divided by the 12 calendar months. Using this as a basis, 1.5 months of funded schedule margin/year = 31.5 working days; 2.0 months = 42 working days; 1 week = 5.25 working days.)

A.4 Subordinate Instrument/Payload Project: An instrument or payload that is managed by a GSFC project office and delivered to its parent GSFC mission flight project.

A.5 Management Reserve: In EVM, the amount of a project's budget, or a contract's total allocated budget, withheld for management control purposes rather than designated for the accomplishment of a specific task or a set of tasks. Contractor management reserve is not part of the project's budget margin.

A.6 Single Project Program: Programs that implement their program objectives and requirements through one of two management approaches: 1) separate program and project structures or 2) a combined structure. These programs tend to have long development and/or operational lifetimes, represent a large investment of Agency resources, and have contributions from multiple organizations/agencies. These programs frequently combine program and project management approaches, which they document through tailoring (see NPR 7120.5E).

A.7 Slack: The amount of duration a task in a network schedule can be delayed without causing a delay to subsequent tasks ("free slack") or the project completion date ("total slack"). Slack is not the same as funded schedule margin.

A.8 Unallocated Future Expenses: The portion of estimated cost required to meet a specified confidence level that cannot yet be allocated to the specific project work breakdown structure sub-elements because the estimate includes probabilistic risks and specific needs that are not

known until these risks are realized. A portion of UFE is held by the project, and another portion of UFE is held by the program office or NASA headquarters mission directorate.

A.9 Critical Path: The sequence of tasks through the integrated master schedule with the least amount of total slack.

A.10 Standalone Instrument/Payload Flight Project: An instrument/payload project that reports directly to a NASA program or a non-GSFC project and is formulated and implemented in accordance with the NASA Project Life Cycle as identified in NPR 7120.5.

A.11 Enterprise Ground System Project: An independent *project* responsible for the development of a large-scale ground system in support of a specific program's ground system requirements. Unlike a mission ground system within a flight project that supports a specific flight project's requirements, the enterprise ground system project reports directly to its parent program.

A.12 Start of Observatory I&T: The start of observatory I&T (the beginning of Interval 2 in this GPR) will vary depending on the nature of the project. It may be defined as: the start of physical integration of the first instrument/payload to the spacecraft bus; the successful completion of the System Integration Review (SIR); the KDP-D approval; the start of pre-instrument integration, inspection, and cleaning of the spacecraft bus; or other suitable event. However defined, the start of observatory I&T should be documented in the project plan and schedule management plan.

Appendix B – Acronyms

CMC	Center Management Council
EVM	Earned Value Management
EVMS	Earned Value Management System
FPD	Flight Projects Directorate
GPR	Goddard Procedural Requirements
GSFC	Goddard Space Flight Center
I&T	Integration and Test
KDP	Key Decision Point
MSR	Monthly Status Review
NASA	National Aeronautics and Space Administration
NPR	NASA Procedural Requirement
NRRS	NASA Records Retention Schedule
ORR	Operational Readiness Review
PDR	Preliminary Design Review
SDR	System Design Review
SMA	Safety and Mission Assurance
SRR	System Requirements Review
TBD	To Be Determined
UFE	Unallocated Future Expenses

Appendix C – Requirements in Table Format

Req't. ID	Requirement	Section Found
7120.7-R001	If it is viewed that the project will remain below the requirements into its next phase, a waiver shall be submitted for CMC concurrence in accordance with GPR 1400.1, Waiver Processing. This applies from formulation through operations. (CANCELLED)	N/A
7120.7-R002	Schedule margin funding shall be calculated according to the highest burn rate appropriate for the phase of the project. (WITHDRAWN, though described in item 2.0f.)	N/A
Funded Schedule Margin for Mission Flight Projects		**2.1**
7120.7-R003	Interval 1: From KDP-C approval (i.e., confirmation) to the start of observatory I&T, 1.5 months of funded schedule margin inserted for every 12 months of Interval 1 planned duration. (REVISED)	2.1a
7120.7-R004	Interval 2: From the start of observatory I&T to delivery of the observatory to the launch site (or to planned storage), 2 months of funded schedule margin inserted for every 12 months of Interval 2 planned duration. (REVISED)	2.1b
7120.7-R005	Interval 3: From observatory delivery to the launch site to launch, 1 week of funded schedule margin inserted for every month of Interval 3 planned duration. (REVISED)	2.1c
Funded Schedule Margin for Standalone Instrument/Payload Flight Projects		**2.2**
7120.7-R017	Interval 1: From KDP-C approval to the start of instrument/payload I&T, 1.5 months of funded schedule margin inserted for every 12 months of Interval 1 planned duration. (The start of instrument/payload I&T varies depending on the nature of the project. Therefore, during the planning phase, projects will define an appropriate milestone or event that constitutes the start of instrument/payload level I&T.) (NEW)	2.2a
7120.7-R018	Interval 2: From the start of instrument/payload I&T to instrument/payload delivery, 2 months of funded schedule margin inserted for every 12 months of Interval 2 planned duration, where instrument/payload delivery may include physical arrival at the launch site or other facility. (NEW)	2.2b
Funded Schedule Margin for Enterprise Ground System Projects		
7120.7-R006	From KDP-C approval (or ground system PDR if there is no enterprise ground system project KDP-C) to ground system ORR, ground system freeze, or other appropriate milestone, 2.0 months of funded schedule margin inserted for every 12 months of planned duration. (REVISED)	2.3a
7120.7-R007	From start of I&T to operational readiness review (ORR): 2 months of funded schedule margin per year. (WITHDRAWN)	N/A

Req't. ID	Requirement	Section Found
7120.7-R008	From ORR to authorization to operate or ground system freeze: 1 week of funded schedule margin per month. (WITHDRAWN)	N/A
Budget Margin for Mission Flight Projects		**3.1**
7120.7-R009	At KDP-A and KDP-B, mission flight projects shall have a minimum budget margin level of 25% with a goal of 30% on cost-to-go through Phase D, and a minimum budget margin level of 15% for Phases E and F. (REVISED)	3.1a
7120.7-R010	At KDP-C, mission flight projects shall have a budget margin level of 25% or higher on cost-to-go through Phase D and 15% for Phases E and F.	3.1b
7120.7-R011	At KDP-D, mission flight projects shall have a budget margin level of 20% or higher on cost-to-go through Phase D and 10% for Phases E and F.	3.1c
7120.7-R012	At the time of delivery to the launch site, mission flight projects shall have a budget margin level of 10% or higher on remaining cost-to-go through Phase D, and 10% for Phases E and F. (REVISED)	31.c
Budget Margin for Standalone Instrument/Payload Flight Projects		**3.2**
7120.7-R019	At KDP-A and KDP-B, standalone instrument/payload flight projects shall have a minimum budget margin level of 25% with a goal of 30% on remaining cost-to-go through Phase D, and 15% for Phases E and F. (NEW)	3.2a
7120.7-R020	At KDP-C, standalone instrument/payload flight projects shall have a budget margin level of 25% or higher on remaining cost-to-go through Phase D, and 15% for Phases E and F. (NEW)	3.2b
7120.7-R021	At KDP-D, standalone instrument/payload flight projects shall have a budget margin level of 20% or higher on remaining cost-to-go through Phase D, and 10% for Phases E and F. (NEW)	3.2c
7120.7-R022	At the time of instrument/payload delivery, standalone instrument/payload flight projects shall have a budget margin level of 10% or higher on remaining cost-to-go through Phase D, and 10% for Phases E and F. (NEW)	3.2d
Budget Margin for Enterprise Ground System Projects		**3.3**
7120.7-R023	At KDP-A and KDP-B (or ground system SRR/SDR if there is no enterprise ground system project KDP-A/KDP-B), enterprise ground system projects shall have a minimum budget margin level of 25% with a goal of 30% on remaining cost-to-go through Phase D (or ground system ORR, ground system freeze, or other appropriatemilestone if no Phase D), and 15% for Phases E and F (or equivalent if no E & F). (NEW)	3.3a
7120.7-R024	At KDP-C (or ground system PDR if there is no enterprise ground system project KDP-C), enterprise ground system projects shall have	3.3b

Req't. ID	Requirement	Section Found
	a budget margin level of 25% or higher on remaining cost-to-go through Phase D (or ground system ORR, ground system freeze, or other appropriate milestone if not Phase D) and 15% for Phases E and F (or equivalent if no E & F). (NEW)	
Monthly Reporting		
7120.7-R013	At the Center's monthly project tag-ups, projects shall present their schedule margin status relative to the required margins. (CANCELLED)	N/A
7120.7-R014	If the schedule margin falls below the agreed-to levels, the presentations shall include the reasons for the shortfall as well as a description of any activities initiated to mitigate the trend. (CANCELLED)	N/A
7120.7-R015	At the Center's monthly project tag-ups, project managers shall present their budget margin status relative to the required levels, (CANCELLED)	N/A
7120.7-R016	If the budget margin falls below the required levels, the presentations shall include the reasons for the shortfall as well as a description of any activities initiated to mitigate the trend. (CANCELLED)	N/A

Appendix D – GPR Applicability Table

Project	7120.7	7120.7A	7120.7B	Notes
ATLAS	X			
ICESat-2	X			
ICON	X			
JWST	X			
LCRD (mission)	X			.7 requirements apply per Code 100 direction
MOMA-MS	X			
Restore-L	X			.7 requirements apply per Code 100 direction
SOC	X			
TESS	X			
XARM		X		
Europa Clipper		X		
GEDI		X		
GOLD		X		
JPSS-2		X		
GOES-S		X		GOES-R Management Control Plan budget margin requirements apply
LandSat 9		X		
XARM		X		
HST Ops			X	Only applies to future major system upgrades
JPSS Ground			X	Only applies to future major system upgrades
JPSS-3			X	
JPSS-4			X	
GOES-R Ground			X	Only applies to future major system upgrades
GOES-T			X	GOES-R Management Control Plan budget margin requirements apply
GOES-U			X	GOES-R Management Control Plan budget margin requirements apply
Lucy			X	
NEN			X	Only applies to future major system upgrades
PACE			X	
Space Network			X	Only applies to future major system upgrades
TSIS-2			X	
WFIRST			X	
Future Missions			X	For missions that submit their proposals in response to Announcements of Opportunity prior to the date of release of GPR 7120.7B, the 7120.7A requirements will apply unless otherwise directed by the CMC.

Future Standalone Instrument/Payload Projects			X	
Future Enterprise Ground System Projects			X	
ESDIS	N/A			May apply to future major system upgrades
ESMO	N/A			May apply to future major system upgrades
HIRMES	N/A			
IXPE	N/A			Follows Marshall Procedural Requirement 7120.1H
LCRD (payload)	N/A			Follow LCRD project requirements
L-Ralph	N/A			Follow Lucy project requirements
LEMNOS	N/A			Compliance with GPR 7120.7 is optional
NIMO	N/A			Not applicable at this time
OCI	N/A			Follow PACE project requirements
POES/MetOp	N/A			Not applicable at this time
PSP	N/A			Follow APL schedule and budget margin guidelines
Resolve	N/A			Follow XARM project requirements
SAR	N/A			
SGSS	N/A			
SSMO	N/A			Not applicable at this time
TDRS	N/A			Not applicable at this time
TIRS-2	N/A			Follow Landsat 9 project requirements

CHANGE HISTORY LOG

Revision	Effective Date	Description of Changes
Baseline	May 4, 2008	Initial Release. This directive replaces GID 7120.1.
Administrative Extension	May 4, 2014	Administratively extended.
Administrative Extension	July 24, 2014	Administratively extended for 1 year.
Administrative Extension	June 16, 2015	Administratively extended for 1 year.
A	February 28, 2017	Updated to current GPR template. Changed title from "Schedule Margins and Budget Reserves to be Used in Planning Flight Projects and Tracking Their Performance" to "Funded Schedule Margins and Budget Margins for Flight Projects." Numbered and indexed requirements ("shall") statements (Appendix C). P.1 and P.2: Clarified. P.8: Added records retention requirements. 1: Rewrote and added new sections on requirements identification, relationship to earned value management, and waivers (rewritten and moved from Section 2). 2: Changed heading to "Funded Schedule Margins." Added header for "Funded Schedule Margins for Observatory Flight Projects," "Funded Schedule Margins for Instrument/Payload Flight Projects," and "Funded Schedule Margins for Enterprise Ground System Projects." Lettered considerations. 2.1: Updated wording of requirements. Rewrote and moved last paragraphs to Section 4 (new). 2.1a: Changed schedule margins for confirmation to the beginning of observatory I&T. 3: Changed "Budget Reserves" to "Budget Margins." 3a: Changed KDP-B requirement from "should" to "shall" and changed margins. Added reserve requirement for KDP-D. 3c: Added new requirement for KDP-D. 4: Added new section for monthly reporting. Appendix A: Added definitions. Appendix B: Added acronyms
B	September 17, 2018	Requirements Cancelled or Withdrawn: R001, R002, R007, R008, R0013, R014, R015 and R016 Requirements Revised: R003, R004, R005, R006, R009 and R012 Requirements Added: R017, R018, R019, R020, R021, R022, R023 and R024

		Title: Changed title from "Schedule and Budget Margins for Flight Projects" to "Funded Schedule Margin and Budget Margin for Flight Projects" P.1: Clarified that the focus of budget and schedule margin requirements is on projects that report to programs. P.2: Added Figure 1 illustrating program and project applicability relationships, and referenced new "GPR Applicability Table" found in Appendix D. P.4: Deleted GPR 1400.1, "Waiver Processing" 1.2: Added definitions for "Funded Schedule Margin" and "Budget Margin;" included new Figure 2 to illustrate budget margin formula 1.3: Added Tables 1, 2 & 3 to summarize funded schedule margin and budget margin requirements for mission projects, instrument/payload projects, and enterprise ground system projects. 1.5: Added new section to address that project noncompliance with GPR requirements will be addressed by the CMC. 2.0 and 2.1 – 2.3: Clarified the requirements for funded schedule margin by mission flight project, standalone instrument/payload flight project, and enterprise ground system project categories; provided additional guidance and supporting illustrations. 2.4: Added section for "Funded Schedule Margin for Other Project Types." 3.0 and 3.1 – 3.3: Clarified the requirements for budget margin by mission flight project, standalone instrument/payload flight project, and enterprise ground system project categories; provided additional guidance and supporting illustrations. 3.4: Added section for "Budget Margin for Other Project Types." 4.0: Eliminated R015 and R016 requirements for reporting margins at Tag-Ups and MSRs and referenced current MSR reporting guidance instead. Appendix A: Added and clarified definitions. Appendix B: Added new acronyms and deleted those no longer referenced. Appendix D: Added appendix for GPR Applicability for current and future projects.
B	August 9, 2023	Administratively extended for 1 year.